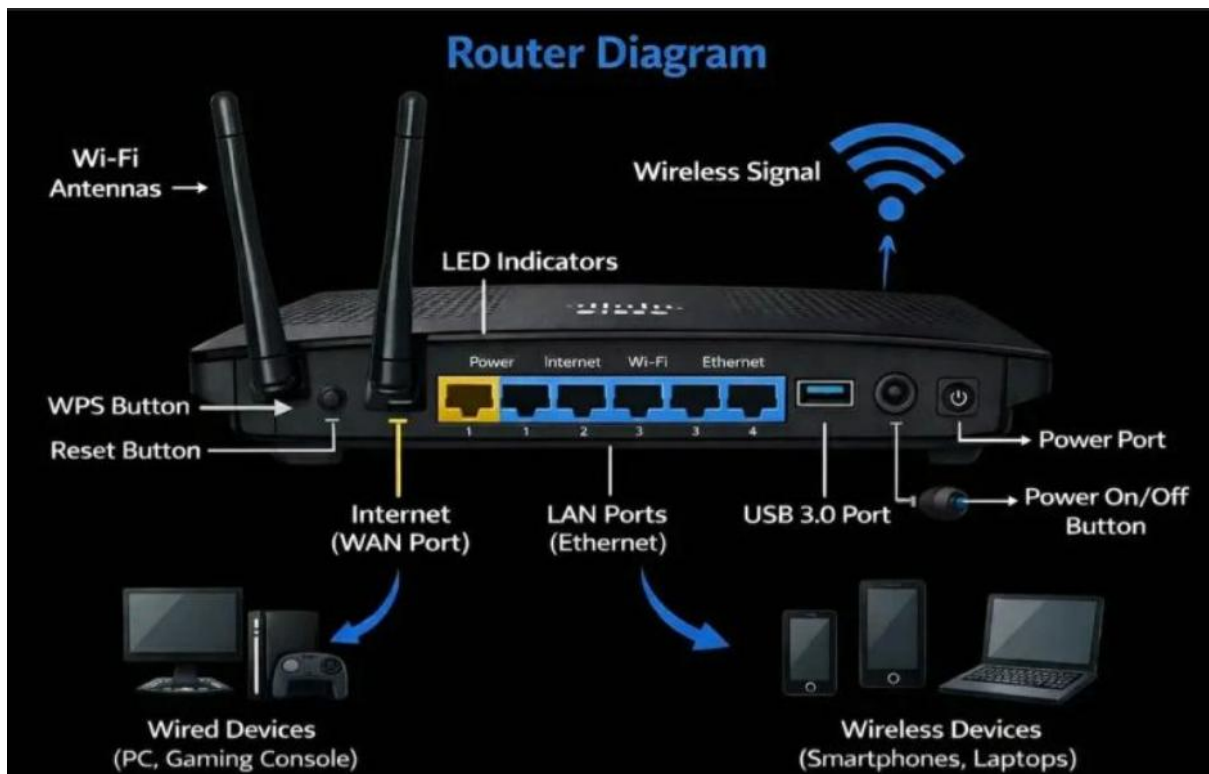


DAY-2

Tasks

1.

Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2.

List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.
Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Feature	Router	Switch
Main function	Connects different networks, such as a home network to the Internet.	Connects multiple devices within the same local network (LAN).
Address used	Mainly uses IP addresses to decide where packets should go.	Mainly uses MAC addresses to forward Ethernet frames to the correct device.
Type of networks	Connects different networks/subnets.	Primarily connects devices within the same LAN.

3.

Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

WIFI-ROUTER



Part	Position	Purpose
Power port	Usually at the back	Connects the router to its power adapter.
WAN/Internet port	Usually at the back; often separated from LAN ports	Connects the router to the modem/ISP connection.
LAN ports	Usually grouped beside the WAN port	Connect wired devices such as PCs, TVs or switches.
Status LEDs	Usually on the front/top	Show power, Internet, Wi-Fi and LAN activity/status.

NETWORK SWITCH



Part	Position	Purpose
Power port	Usually on the rear/side	Connects the switch to its power supply.
Ethernet Port 1	Front/rear depending on model	Connects the switch to the router or another network device.
Ethernet Port 2	Next to the other Ethernet ports	Connects a PC or another wired device.
Port LEDs	Usually above/below each Ethernet port	Indicate whether a connection is active and/or data is being transmitted.

4.

Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

First, I would connect the ISP line to the modem/ONT and then connect the modem/ONT to the WAN/Internet port of the router.

Then I would connect one of the router's LAN ports to the network switch using an Ethernet cable.

The 8 gaming PCs would then be connected to 8 Ethernet ports on the switch.

The router should come first because it connects the café's local network to the ISP/Internet and handles routing between the café network and external networks.

The switch is then connected to the router because it provides enough Ethernet ports for all 8 gaming PCs. A switch is ideal for connecting the PCs within the same local network and allows them to communicate efficiently.

So, in one line:

ISP → Modem → Router → Switch → 8 Gaming PCs

That setup gives the café Internet connectivity while allowing all 8 PCs to communicate over the local network.